

PROFESSIONAL NURSING COACH

PATHOLOGY I

PNC ULTRA-LMR HYBRID MASTER COMPENDIUM • NUR-09 • VERIFIED

General + special + lab/bank • Batches 1–3 • 31 LMR pages + cover & TOC • NORCET / RRB / ESIC / DSSSB

Batch	Units	Pp
B1	I: language, injury, necrosis, adaptations, inflammation, healing, neoplasia, clots, edema	10
B2	II-A: lung, COPD, vascular, RHD, PUD, IBD, liver, gallbladder, bone, DM	10
B3	II-B + III: tumors, RA/OA, goitre, CBC, coag, bank, chemistry, reactions	11

WhatsApp DM: +91 90571 22020 • Hybrid: navy hero + chips + teal bars + Trap/Hit + Pearl + Speed-check • v1.0 Sep-2026

Pp	Block	Focus
4-13	I (B1)	Language, injury, necrosis, adaptations, inflammation, healing, neoplasia, clots, edema
14-23	II-A (B2)	Lung, COPD, vascular, RHD, PUD, IBD, liver, gallbladder, bone, DM
24-34	II-B+III (B3)	Tumors, RA/OA, goitre, CBC, coag, bank, chemistry, plasma, reactions

How to use: hero first, chips glance, table study, trap vs hit, pearl anchor, speed-check close. Double-QA pending sign-off on doses/thresholds per current guidance.

Q	Standard	Status
Q1	Units I-III mapped, batches 1-3	Done
Q2	Lesion names + triads match standard pathology	Needs sign-off
Q3	Lab cutoffs + transfusion rules current	Needs sign-off
Q4	One concept/page, fonts 10.5-21pt, 30% air	Done

Etiology → pathogenesis → morphology → clinical! Lesion words (hyperplasia vs hypertrophy vs atrophy vs metaplasia vs dysplasia vs neoplasia) + biopsy + cytology + autopsy + fixatives — speak lesion, think mechanism!

**Etiology → patho →
morpho → clinic**

**-plasia words
decide benign**

**Biopsy + cyto +
autopsy + fix!**

★ WORD — MEANS

Word	Means
Hyperplasia/hypertrophy	More cells / bigger cells (reversible)
Atrophy/metaplasia	Shrink / one adult-to-another (smoke!)
Dysplasia/neoplasia	Disordered / new-clonal (benign vs malign)
Biopsy/cyto/autopsy	Tissue/cells/death-cause + fixatives

⚠ FATAL TRAP

Calling metaplasia 'cancer' + skipping biopsy + 'clinical enough' = wrong + suit! Morphology proves + grades + stages + guides + consent + registry.

★ NORCET HIT

HIT: Barrett (squamous-to-columnar) + cervical + bronchial metaplasia = surveillance! Dysplasia low → high → in-situ ladder!

★ PNC PEARL: Adapt, disarrange, or go clonal — name it right.

► Smoker + bronchial squamous change. Name + act?

Metaplasia — stop-smoke + surveillance + follow-up.

CELL INJURY: REVERSIBLE VS DEATH

Hypoxia + toxins + free-radicals + membrane + mitochondria! Swelling + fatty (reversible) vs necrosis/apoptosis (dead) — Na/K-pump fail + calcium flood + blebs + rupture = point of no return!

**Swell + fatty
= reversible**

**Pump-fail + Ca
= death**

**Hypoxia + toxin
+ radical!**

▲ STAGE — CLUE — FATE

Stage	Clue	Fate
Swelling	Na/K fail + water + blebs	Reversible + rest
Fatty	Triglyceride + liver + alcohol	Reversible + abstain
Necrosis	Rupture + inflammation + enzymes	Dead + scar
Apoptosis	Shrink + bodies + no-inflammation	Programmed + quiet

⚠ FATAL TRAP

Re-perfusing + 'more O2' + missed radicals + no antioxidants-plan = reperfusion injury! Stagger + control + cause + support + HDU + follow-up + registry.

★ NORCET HIT

HIT: Karyolysis + pyknosis + karyorrhexis = nuclear death trio! Coagulative (heart/kidney) vs liquefactive (brain) vs caseous (TB) vs fat (pancreas)!

★ PNC PEARL: Blebs burst, nuclei crumble — death signed.

► Swollen + blebs + intact membrane. Fate + act?

Reversible — O2 + cause + rest + review.

NECROSIS + GANGRENE

Coagulative (heart/kidney/spleen) + liquefactive (brain/abscess) + caseous (TB!) + fat (pancreas/chalk) + fibrinoid (vessels)! Dry (mummified + line) vs wet (infected + spread) vs gas (clostridial + crepitus + emergency)!

Coag/lique/caseous /fat/fibrinoid	Dry = line; wet/gas = OT!	Debride + antibiotics + OT!
--------------------------------------	------------------------------	--------------------------------

◆ TYPE — CLUE — ACT

Type	Clue	Act
Dry	Mummified + line + arterial	Debride + revascularize + review
Wet	Moist + infected + spread	Debride + antibiotics + OT + ICU
Gas	Crepitus + toxin + shock	OT + penicillin + ICU + HDU
Internal	Bowel/pancreas + pain + shock	NPO + OT + ICU + HDU

⚠ FATAL TRAP

Dressing-only + 'dry will heal' + missed wet/gas + no OT + diabetic-foot-neglect = sepsis + limb + death!
Probe + X-ray + culture + sugar + vascular + OT + rehab + footwear.

★ NORCET HIT

HIT: Fournier + Meleney + diabetic + pressure + venous + arterial + neuropathic + offload + graft + footwear + registry!

★ PNC PEARL: Line limits, spread operates, gas gallops.

► Crepitus + shock + foul + diabetic foot. Act trio?

Gas — OT + penicillin + ICU + sugar + blood.

Atrophy (disuse/denervation/ischemia) + hypertrophy (muscle/heart!) + hyperplasia (prostate/endo!) + metaplasia (smoke/stone!) + dysplasia (cervix/oral!) — physiologic vs pathologic decides watch vs workup!

**Atrophy = shrink
(disuse)**

**Hyper = more/big
(load/hormone)**

**Meta/dysplasia =
watch/workup!**

Σ ADAPT — EXAMPLE — ACT

Adapt	Example	Act
Atrophy	Disuse + denervation + ischemia	Physio + cause + review
Hypertrophy	Heart + muscle + uterus	Load + BP + echo + rest
Hyperplasia	Prostate + endometrial	Hormones + USG + biopsy
Meta/dysplasia	Smoke/stone/cervix	Stop + scope + grade

⚠ FATAL TRAP

'Bigger means stronger' + missed pathologic + no echo/biopsy + 'supplement it' = failure + cancer + death! Cause + image + grade + treat + follow-up + registry.

★ NORCET HIT

HIT: LVH + ECG + echo + BP + athlete vs aortic + compliance + drugs + surgery + sport + genes! BPH + PSA + USG + flow + meds + TURP!

★ PNC PEARL: Adapt asks why — answer with workup.

► **Smoker + bronchial metaplasia + dysplasia-low. Plan?**

Stop + scope + grade + surveillance + date.

Rubor + tumor + calor + dolor + functio! Vascular (dilatation + permeability) + cellular (margination + roll + pave + emigrate + chemotaxis + phagocytose!) + mediators (histamine + PG + complement + kinins + clot)!

5 cardinals +
loss-function

Roll-pave-emigrate
-eat!

Histamine + PG +
complement!

★ STEP — EVENT

Step	Event
Vascular	Dilatation + permeability + exudate + edema
Cellular	Roll + pave + emigrate + chemotaxis + eat
Mediators	Histamine + PG + complement + kinins + clot
Outcomes	Resolve/suppurate/chronic/scar + fever + CRP

⚠ FATAL TRAP

Steroids/NSAIDs-mask + missed pus + 'inflammation = infection' + no culture/drain = abscess + sepsis + death!
Cause + culture + drain + antibiotics + support + HDU.

★ NORCET HIT

HIT: Exudate (protein + cells) vs transudate (pressure!)
+ Light + Rivalta + LDH + count + culture + cytology!

★ PNC PEARL: Dilate, leak, eat — then resolve or scar.

► **Hot + red + pus + fever + tender + fluctuant. Act trio?**

Abscess — culture + drain + antibiotics.

CHRONIC + GRANULOMATOUS INFLAMMATION

Macrophage + lymphocyte + plasma + giant + fibrosis + granuloma! TB (caseating + Langhans + AFB!) + sarcoid (non-caseating + ACE!) + leprosy + syphilis + fungal + foreign-body + Crohn + silence + steroids + ATT + follow-up!

Macro + giant +
fibrosis + granuloma

TB caseating +
Langhans + AFB

Sarcoid non-case +
ACE + steroids!

▲ TYPE — CLUE — ACT

Type	Clue	Act
TB	Caseating + Langhans + AFB + night-sweats	ATT + DOT + contacts + nutrition
Sarcoid	Non-caseating + ACE + hilar + eye/skin	Steroids + eye + lung + review
Leprosy/syphilis	Nerve/chancere + serology	MDT/penicillin + deformity + counsel
Foreign/Crohn	Polarizable + skip + cobble	Remove + steroids + surgery

⚠ FATAL TRAP

Steroids-without-TB-exclusion + 'sarcoid always' + missed AFB/HIV + no DOT = disseminated + dead!
Biopsy + AFB + CBNAAT + HIV + eye + lung + senior + registry.

★ NORCET HIT

HIT: Epithelioid + Langhans (horse-shoe!) + Schaumann + asteroid + multinucleate + caseate + stain + culture + PCR!

★ PNC PEARL: Granuloma guesses, AFB proves, DOT cures.

► Non-caseating + hilar + ACE-high + eye-red. Name + act?

Sarcoid — steroids + eye + lung + review.

Hemostasis → inflammation → proliferation (fibro + angio + granulation!) → remodeling (collagen + strength!)! Primary (clean + sutured) vs secondary (gap + granulate) vs tertiary (delayed!) + dehiscence + keloid + ulcer factors!

**Clot → clean →
granulate → remodel**

**Primary/secondary/
tertiary intents**

**Dehiscence + keloid
+ ulcer watch!**

◆ PHASE — EVENT — CARE

Phase	Event	Care
Hemostasis	Clot + platelets + fibrin	Press + TXA + Hb + review
Inflammation	Clean + neutrophils + exudate	Debride + antibiotics + dress
Proliferation	Fibro + angio + pink granulation	Moist + nutrition + O2 + offload
Remodeling	Collagen + strength + scar + review	Protect + physio + scar + date

⚠ FATAL TRAP

Tight-sutures + infection + diabetes + steroids + smoking + 'healing by itself' = dehiscence + evisceration + death! Tension-free + sterile + sugar + protein + O2 + offload + VAC + graft.

★ NORCET HIT

HIT: Keloid (beyond!) vs hypertrophic (within!) + pressure + silicone + steroid + laser + surgery! Ulcer: venous + arterial + neuropathic + pressure + offload + dress!

★ PNC PEARL: Moist, fed, oxygenated, offloaded — healing's four.

► Burst + bowel out + post-op day-5 + shock. Act?

Evisceration — cover + NPO + OT + blood + ICU.

Benign (capsule + slow + no-spread + -oma!) vs malignant (invade + mets + anaplasia + grading + staging TNM!!) Dysplasia → in-situ → invasive + paraneoplastic + cachexia + clot + biopsy + MDT + cure + palliative!

**Benign: capsule +
slow + -oma**

**Malignant: invade +
mets + grade/stage**

**Biopsy + MDT +
cure + palliate!**

Σ FEATURE — BENIGN — MALIGNANT

Feature	Benign	Malignant
Growth	Slow + capsule + push	Fast + invade + mets
Cells	Well-diff + uniform	Anaplastic + mitotic + N/C-high
Suffix	-oma (fibro/lip/myo)	Carcinoma/sarcoma/blastoma
Act	Excise + histology	Stage + MDT + cure + palliate

⚠ FATAL TRAP

'Benign, ignore' + no histology + missed red (bleed/weight/lump/node) + quack-cut = upstaged + lost! Every lump: exam + image + biopsy + stage + MDT + consent + registry.

★ NORCET HIT

HIT: Paraneoplastic (hyperCa + SIADH + Cushing + Trousseau + acanthosis) + cachexia + clot + anemia + fever + night-sweats + club + nodes!

★ PNC PEARL: Grade tells temper, stage tells spread.

► **Hard + fixed + nodes + weight-loss + 60 yrs. Ladder?**

Biopsy + stage + MDT + consent + plan.

THROMBOSIS, EMBOLISM + SHOCK (PATH)

Virchow: stasis + injury + hypercoagulable! Pale (arterial) vs red (venous) + lines-of-Zahn + fate (lyse/organize/embolize!) + PE + DVT + DIC + fat/air/amniotic + shock (hypo/cardiogenic/distributive/obstructive)!

Virchow 3 + Zahn lines	Pale = arterial; red = venous	PE/DVT/DIC + shock hunt!
---------------------------	----------------------------------	-----------------------------

★ CLOT – CLUE – ACT

Clot	Clue	Act
DVT	Pain + swell + Homan? + Doppler	No-massage + heparin + stockings
PE	Tachy + dyspnea + ECG + CT	O2 + heparin + ICU + lysis-select
DIC	Bleed + clot + labs + trigger	Products + cause + ICU + HDU
Shock	Cold/warm + lactate + urine	Fluids + pressors + cause + ICU

⚠ FATAL TRAP

Massage + 'leg pain normal' + missed PE/DIC + heparin-in-bleed + no cause-hunt = embolus + bleed + death! Doppler + CT + D-dimer + score + anticoag + filter + audit.

★ NORCET HIT

HIT: Fat (fracture + petechiae + confusion) + air (lines + neuro) + amniotic (labour + DIC) + septic + anaphylaxis + neurogenic + labs + registry!

★ PNC PEARL: Virchow whispers, embolus shouts.

► Calf swell + tachy + dyspnea + pleuritic. Lane?

DVT + PE — Doppler + CT + heparin + ICU.

Forces: hydrostatic vs oncotic + permeability + lymph + Na-retention! Pitting + dependent + anasarca + pulmonary + cerebral + angio! Transudate (low-protein, cirrhosis/CCF/nephrotic) vs exudate (high-protein, infection/malignancy/TB)!

Push vs pull + leak + lymph	Transudate = pressure/low-protein	Exudate = leak/infect/malig!
--------------------------------	--------------------------------------	---------------------------------

▲ FLUID — CLUE — ACT

Fluid	Clue	Act
Transudate	Low-protein + cirrhosis/CCF/nephrotic	Salt + diuretics + cause + review
Exudate	High-protein + fever + TB/malig	Tap + culture + cytology + treat
Pulmonary	Crackles + O2-drop + X-ray	Sit + O2 + diuretics + cause + ICU
Angio/cerebral	Swell + stridor/coma	Adrenaline/mannitol + airway + ICU

⚠ FATAL TRAP

Diuretics-without-cause + missed
exudate/TB/malignancy + 'swelling normal' + no tap =
masked + dead! Tap + Light + protein/LDH + culture +
cytology + ADA + senior + review.

★ NORCET HIT

HIT: Light's + Rivalta + SAAG + cell-count + glucose +
amylase + BNP + JVP + weight + I/O + K + senior +
audit!

★ PNC PEARL: Tap it, type it, treat cause.

► Fever + effusion + lymphocytic + ADA-high. Name + act?

TB — ATT + steroids-select + follow-up + contacts.

PNEUMONIA, ABSCESS + PULMONARY TB

Lobar (rusty + air-bronchogram!) vs broncho (patchy) vs abscess (cavity + air-fluid + foul!) vs TB (upper + cavity + granuloma + AFB + DOT!) — X-ray + sputum + CBNAAT + antibiotics + drain + ATT + contacts + follow-up!

**Lobar rusty +
air-bronchogram**

**Abscess cavity +
foul + drain**

**TB upper + AFB
+ DOT!**

★ LESION — CLUE — ACT

Lesion	Clue	Act
Lobar	Rusty + air-bronchogram + chill	Antibiotics + O2 + fluids + review
Abscess	Cavity + air-fluid + foul + club	Drain + antibiotics + physio + OT
TB	Upper + cavity + granuloma + AFB	ATT + DOT + contacts + nutrition
All	X-ray + sputum + CBNAAT + culture + HDU	

⚠ FATAL TRAP

Steroids-first + 'viral' + missed AFB/abscess + no DOT/contacts + early-discharge = MDR + rupture + death! Sputum + CBNAAT + HIV + X-ray + drain + ATT + audit.

★ NORCET HIT

HIT: Ghon + Ranke + miliary + effusion + lymph + spine + meningitis + DOTS + Nikshay + DBT + BCG + audit!

★ PNC PEARL: Rusty, foul, upper — three doors, three keys.

► **Cavity + air-fluid + foul + club + fever. Name + act?**

Abscess — drain + antibiotics + physio + OT.

COPD: BRONCHITIS, EMPHYSEMA, ASTHMA, BRONCHIECTASIS

Blue-bloater (bronchitis + cough + cyanosis!) vs pink-puffer (emphysema + barrel + pursed!) + asthma (reversible + triggers!) + bronchiectasis (railroad + foul + club + HRCT + postural!) — smoke + O2 + inhalers + vaccines + rehab + surgery!

Blue-bloater vs
pink-puffer

Asthma reversible
+ triggers

Bronchiectasis: rail
+ foul + club!

▲ DISEASE — CLUE — ACT

Disease	Clue	Act
Bronchitis	Cough + cyanosis + CO2 + cor	O2 + bronchodilator + rehab + vax
Emphysema	Barrel + pursed + bust + AAT	O2 + rehab + surgery-select + genes
Asthma	Reversible + triggers + family	SABA + ICS + plan + technique
Bronchiectasis	Rail + foul + club + HRCT	Drain + antibiotics + surgery + vax

⚠ FATAL TRAP

O2-flood + missed CO2 + 'COPD all' + no inhaler-technique + smoke-continue = narcosis + death + readmit! Targeted-O2 + ABG + technique + quit + vaccines + rehab + NIV + ICU.

★ NORCET HIT

HIT: Reid + barrel + Hoover + pursed + accessory + tripod + ABG + cor + polycythemia + transplant + genes + audit!

★ PNC PEARL: Bloaters blue, puffers pink, plans personal.

► Barrel + pursed + bust + smoker + CO2-high. Name + O2?

Emphysema — targeted O2 + ABG + rehab + quit.

ATHEROSCLEROSIS, ISCHEMIA + INFARCTION

Fatty-streak → plaque → rupture → clot → infarct! LDL + HTN + smoke + DM + age + family + CRP + calcium-score + angio + statin + antiplatelet + stent/bypass + rehab + diet + quit + audit!

Streak→plaque→
rupture→clot

LDL + HTN +
smoke + DM

Statin + platelets
+ stent/bypass!

◆ STAGE — ACT

Stage	Act
Risk	Lipids + BP + sugar + smoke + diet + move
Angina	ECG + troponin + aspirin + nitro + cath
Infarct	MONA + PCI + lysis-window + ICU + rehab
After	DAPT + statin + rehab + quit + genes

⚠ FATAL TRAP

Indigestion + 'gas' + missed ECG/troponin + NSAID + delay-PCI = dead muscle + dead patient! ECG-10 + troponin + MONA + cath + window + ICU + rehab + audit.

★ NORCET HIT

HIT: Coagulative + wavy + neutrophils + granulation + scar + weeks + rupture + aneurysm + mural + Dressler + rehab!

★ PNC PEARL: Time is muscle — ECG in ten.

► **Crushing + sweat + radiation + ECG-high + troponin+. Act?**

Infarct — MONA + PCI + ICU + rehab.

Rheumatic (Aschoff + Anitschkow + Carey-Coombs + valves!) + vegetations + Osler/Roth/Janeway/splinter + blood-cultures + echo + penicillin + surgery + prophylaxis + dental + card! Strep-throat treated = valve saved!

**Aschoff +
vegetations + nodes**

**Cultures + echo
+ penicillin**

**Prophylaxis +
dental + card!**

Σ LESION — ACT

Lesion	Act
RHD valves	Echo + penicillin + surgery + anticoag
IE vegetations	Cultures + echo + IV + surgery + HDU
Complications	Emboli + failure + abscess + watch + ICU
Prevent	Throat + dental + prophylaxis + card + audit

⚠ FATAL TRAP

Fever + murmur + 'viral' + no cultures/echo + missed emboli/abscess + dental-ignored = valve + brain + death! Cultures-first + echo + IV + surgery + prophylaxis + audit.

★ NORCET HIT

HIT: FROM-J (fever + Roth + Osler + murmur + Janeway!) + splinter + club + spleen + urine + anemia + ESR + CRP + rheumatoid + HACEK!

★ PNC PEARL: Fever + murmur = cultures + echo, always.

► **Fever + murmur + Osler + IV-drug + echo-mass. Name + act?**

IE — cultures + IV + echo + surgery + HDU.

H. pylori + NSAID + acid + ulcer + bleed + perforation + UBT + triple + scope + PPI + quit!
Leukoplakia (tobacco-white + biopsy + dysplasia!) vs lichen vs thrush(scrape!) vs hairy (EBV/HIV!) — white needs biopsy, always!

**H. pylori + NSAID
+ scope + triple**

**White + tobacco
= biopsy**

**Scrape thrush;
hairy = HIV!**

★ LESION — ACT

Lesion	Act
PUD	UBT + scope + triple + PPI + quit + bleed-watch
Leukoplakia	Biopsy + dysplasia + stop + laser + review
Thrush/hairy	Scrape + antifungal/HIV + ART + follow
Red	Perforation + bleed + OT + HDU + ICU

⚠ FATAL TRAP

NSAID-continue + 'acidity' + missed bleed/perforation + tobacco-white-ignored = shock + cancer + death!
Scope + biopsy + eradicate + quit + bleed-card + surgery + audit.

★ NORCET HIT

HIT: Curling (burns!) + Cushing (brain!) + NSAID + Zollinger (gastrin + diarrhea + MEN!) + vagotomy + PPI + H2 + antacid!

★ PNC PEARL: Scope bleeds, biopsy whites.

► NSAID + black stools + shock + rigid. Lane?

Bleed + perforation — scope + OT + blood + ICU.

TYPHOID, CROHN + ULCERATIVE COLITIS

Typhoid (rose + step-fever + bradycardia + Widal/culture + ceftriaxone!) + Crohn (skip + cobble + transmural + fistula + B12!) + UC (continuous + rectum-up + bloody + toxic-megacolon!) — scope + biopsy + ASA/steroids/biologics + surgery!

**Rose + step +
bradycardia**

**Skip + cobble
= Crohn**

**Continuous + bloody
= UC!**

▲ DISEASE — CLUE — ACT

Disease	Clue	Act
Typhoid	Rose + step + bradycardia + culture	Ceftriaxone + fluids + vaccine + WASH
Crohn	Skip + cobble + transmural + fistula	Steroids/biologics + B12 + surgery
UC	Continuous + bloody + toxic-mega	ASA/steroids + scope + colectomy

⚠ FATAL TRAP

Steroids-without-TB-exclusion + 'IBS' + missed toxic-megacolon/perforation + no scope/biopsy = rupture + death + cancer! Scope + biopsy + severity + MDT + surgery + surveillance.

★ NORCET HIT

HIT: Backwash + pseudopolyps + lead-pipe + thumbprint + toxic + cancer + PSC + joints + eyes + skin + pyoderma + uveitis!

★ PNC PEARL: Skip vs continuous — scope settles.

► **Bloody + continuous + toxic + dilated + fever. Name + act?**

Toxic megacolon (UC) — NPO + steroids + surgery + ICU.

Viral (A/E-water + B/C/D-blood + serology + vaccine!) + amoebic/pyogenic (anchovy + pigtail!) + cirrhosis (fibrosis + portal-HTN + varices + ascites + encephalopathy + MELD + band/TIPS/transplant!)

A/E-water +
B/C/D-blood

Anchovy + pigtail
for abscess

Fibrosis + varices
+ MELD!

◆ LIVER – ACT

Liver	Act
Viral	Serology + vaccine + tenofovir/DAA + contacts
Abscess	Anchovy + pigtail + metronidazole + senior
Cirrhosis	Band + diuretics + lactulose + TIPS + transplant
Failure	Encephalopathy + coag + ICU + transplant + HDU

⚠ FATAL TRAP

Alcohol-continue + paracetamol-flood + missed varices/encephalopathy + no vaccine/contacts + 'liver-tonic' = bleed + coma + death! Band + lactulose + vaccine + abstain + transplant + audit.

★ NORCET HIT

HIT: Spider + palmar + gynecomastia + caput + spleen + ascites + flapping + fetor + SAAG + MELD + Child-Pugh + HCC-screen!

★ PNC PEARL: Serology sorts, vaccine shields, band bleeds-stops.

► Vomit-blood + cirrhosis + shock + melena. Act trio?

Variceal — band + octreotide + blood + ICU.

Murphy + fat-forty-fertile + USG + antibiotics + lap-chole + ERCP + stone + diet! Pancreatitis (epigastric-belt + amylase/lipase + gallstone/alcohol + Ranson + necrosis + pseudocyst + ICU + NPO + fluids + nutrition)!

**Murphy + fat-forty
+ USG + lap**

**Belt-pain +
amylase/lipase**

**Ranson + necrosis
+ ICU!**

Σ ORGAN — ACT

Organ	Act
Gallbladder	USG + antibiotics + lap-chole + ERCP + diet
Pancreas-mild	NPO + fluids + analgesia + cause + review
Necrosis	CT + antibiotics + drain + OT + ICU + HDU
Chronic	Enzymes + DM + pain + surgery + abstain

⚠ FATAL TRAP

Fatty-continue + alcohol-continue + missed cholangitis/necrosis + no ERCP/drain + early-discharge = sepsis + death + diabetes! Charcot + Reynolds + Ranson + BISAP + senior + HDU.

★ NORCET HIT

HIT: Courvoisier (palpable + painless + malignancy!) + Grey-Turner + Cullen + steatorrhea + calcify + pseudocyst + fistula + audit!

★ PNC PEARL: Murphy points, lipase proves, necrosis operates.

► **Belt-pain + lipase-high + shock + calcium-low. Grade + lane?**

Severe — CT + ICU + NPO + fluids + nutrition.

Hematoma → callus → remodel + Sequestrum (dead!) + involucrum (new!) + cloaca (pus-hole!) + Brodie + X-ray + culture + antibiotics + debride + fix + graft + physio + follow-up + audit!

Callus + remodel + graft	Sequestrum dead + involucrum new	Culture + debride + fix + physio!
-------------------------------------	---	--

★ ITEM — RULE

Item	Rule
Healing	Hematoma + callus + remodel + nutrition + fix
Osteomyelitis	Sequestrum + involucrum + cloaca + culture
Treat	Antibiotics + debride + fix + graft + physio
After	Follow-up + growth + deformity + review

⚠ FATAL TRAP

Antibiotics-only + missed sequestrum + 'healed' + no debride/fix + lost follow-up = chronic + sinus + deformity + amputation! X-ray + ESR/CRP + culture + OT + graft + audit.

★ NORCET HIT

HIT: Sickle + TB (Pott + gibbus!) + diabetic-foot + pressure + prosthesis + biofilm + VAC + beads + hyperbaric + registry!

★ PNC PEARL: Dead out, new in, pus drained.

► Discharging sinus + sequestrum + fever + child. Act trio?

Chronic — culture + debride + antibiotics + fix.

Autoimmune-beta (T1!) + resistance (T2!) + MODY + gestational + HbA1c + micro (eye/kidney/nerve!) + macro (heart/brain/leg!) + DKA/HHS + foot + diet + drugs + insulin + pump + eyes/kidney/feet + audit!

**T1-auto + T2-resist
+ MODY + GDM**

**Micro: eye/kidney/
nerve + macro**

**HbA1c + foot +
screen yearly!**

▲ ITEM — RULE

Item	Rule
Diagnose	Sugar + HbA1c + autoantibodies + C-peptide
DKA/HHS	Fluids + insulin + K + cause + ICU + HDU
Chronic	Eyes + kidney + feet + heart + nerves + audit
Live	Diet + move + drugs + pump + school + camp

⚠ FATAL TRAP

Missed DKA + insulin-stop + 'sugar controlled' + no foot/eye/kidney screen + no sick-rules = coma + blind + dialysis + chop! HbA1c + screens + kit + school + 24-h + registry.

★ NORCET HIT

HIT: Kimmelstiel + Armanni + necrobiosis + cheiro + frozen + Charcot + amyotrophy + gastroparesis + cystopathy + audit!

★ PNC PEARL: A1c + eyes + feet + kidneys — yearly quartet.

► Fruity + Kussmaul + dehydrated + sugar-high + child. Act?

DKA — fluids + insulin + K + ICU + cause.

LUNG + ESOPHAGEAL/GASTRIC TUMORS

Smoker + cough + hemoptysis + club + hilar + biopsy + stage + MDT! Esophageal (dysphagia + scope + bougie + stent!) + gastric (ulcer-cancer + scope + H. pylori + surgery + chemo!) — red + weight + nodes = scope + biopsy, always!

**Smoker + cough +
hemoptysis = scope**

**Dysphagia +
scope + stent**

**Ulcer-cancer +
H. pylori hunt!**

★ TUMOR — ACT

Tumor	Act
Lung	X-ray + CT + biopsy + stage + MDT + quit
Esophageal	Scope + bougie + stent + chemo/radio
Gastric	Scope + H. pylori + surgery + chemo
All	Nodes + mets + nutrition + palliative + audit

⚠ FATAL TRAP

'Smoker's cough' + antacids-months + no scope/biopsy + missed dysphagia/weight = stage-IV + dead! Scope + biopsy + stage + MDT + quit + nutrition + palliative.

★ NORCET HIT

HIT: Pancoast (Horner + arm-pain!) + SVC + paraneoplastic + club + nodes + mets + linitis-plastica + Krukenberg + audit!

★ PNC PEARL: Smoke + swallow-trouble + weight-loss = scope now.

► Dysphagia + weight-loss + smoker + nodes. Ladder?

Scope + biopsy + stage + MDT + quit + support.

COLORECTAL + LIVER/GALLBLADDER/PANCREAS TUMORS

Bleed + change + mass + CEA + scope + polyp + surgery + chemo! Liver (AFP + HBV + cirrhosis + triple-phase + transplant!) + gallbladder (stones + porcelain + radical!) + pancreas (painless-jaundice + Whipple + CA19!)!

**Bleed + change
= scope + polyp**

**AFP + HBV +
triple-phase**

**Painless-jaundice
= Whipple!**

▲ TUMOR — ACT

Tumor	Act
Colorectal	Scope + polyp + CEA + surgery + chemo
Liver	AFP + triple-phase + transplant + TACE
Gallbladder	Stones + porcelain + radical + nodes
Pancreas	CA19 + Whipple + chemo + stent + pain

⚠ FATAL TRAP

Piles-forever + 'gastritis' + missed scope/AFP/painless-jaundice + no staging = unresectable + dead! Scope + markers + stage + MDT + surgery + chemo + palliative.

★ NORCET HIT

HIT: FAP + HNPCC + UC + polyp-cancer + Courvoisier + Trousseau + Sister-Joseph + Virchow + audit!

★ PNC PEARL: Bleed scoped, markers mapped, jaundice whipped.

► Painless-jaundice + mass + weight-loss + 60 yrs. Suspect + act?

Pancreatic — CA19 + stage + Whipple + MDT.

BONE TUMORS (LOW-YIELD MAP)

Night-pain + swelling + X-ray (sunburst/onion/codman!) + biopsy + stage + limb-salvage + chemo + mets-lungs + follow-up! Osteosarcoma (knee + teens + sunburst!) + Ewing (diaphysis + onion + EWS!) + myeloma (elderly + lytic + Bence-Jones!)

Night-pain +
swelling = X-ray

Sunburst/onion/
Codman signs

Biopsy + salvage
+ chemo!

◆ TUMOR — CLUE — ACT

Tumor	Clue	Act
Osteosarcoma	Knee + teens + sunburst	Biopsy + salvage + chemo
Ewing	Diaphysis + onion + fever	Biopsy + chemo/radio + surgery
Myeloma	Elderly + lytic + Bence-Jones	Chemo + transplant + bones

⚠ FATAL TRAP

'Growing pains' + massage + missed X-ray/biopsy + quack-plaster = mets + limb-lost + dead! X-ray + MRI + biopsy + stage + MDT + limb-salvage + chemo + physio.

★ NORCET HIT

HIT: Giant-cell (soap-bubble + egg-shell!) + chondrosarcoma + metastasis (prostate/lung/breast/kidney/thyroid!) + ALP + audit!

★ PNC PEARL: Night bone pain earns daylight X-ray.

► Teen + knee swelling + sunburst + night-pain. Ladder?

Biopsy + stage + salvage + chemo + MDT.

RHEUMATOID VS OSTEOARTHRITIS

RA (small + symmetric + morning-stiffness + RF/anti-CCP + erosions + nodules + lungs + MTX + biologics!) vs OA (big + asymmetric + use-pain + osteophytes + Heberden/Bouchard + weight + physio + replace!)

RA: small + symmetric
+ morning + RF

OA: big + use-pain
+ osteophytes

MTX/biologics vs
weight/replace!

Σ FEATURE — RA — OA

Feature	RA	OA
Joints	Small + symmetric	Big + asymmetric + spine/hip/knee
Stiffness	Morning-long + improves	Use-pain + rest-eases
Labs/X-ray	RF/CCP + erosions	Osteophytes + Heberden + space-loss
Treat	MTX + biologics + steroids	Weight + physio + replace

⚠ FATAL TRAP

Steroids-only + missed MTX-window + 'age-pain' + no X-ray/labs + quack-oils = deformity + disability + dead!
Early + DMARD + biologics + physio + surgery + vaccines + audit.

★ NORCET HIT

HIT: Felty + Caplan + Sjogren + nodules + vasculitis + lung + heart + eyes + TB-screen + biologics + pregnancy + audit!

★ PNC PEARL: Small-symmetric-morning = RA realms.

► Small + symmetric + morning-long + RF+. Name + drug?

RA — MTX + biologics + steroids + physio.

Iodine + TSH + diffuse/multinodular + toxic + eye + surgery + cancer (papillary + medullary + calcitonin + neck-nodes + thyroidectomy + radioiodine + thyroxine + follow-up)! Cold + solid + hoarse + nodes = biopsy!

Iodine + TSH +
diffuse/nodular

Cold + hoarse +
nodes = biopsy

Surgery + iodine +
thyroxine + date!

★ TYPE — ACT

Type	Act
Diffuse/toxic	Iodine + carbimazole + surgery + eye + review
Nodular	USG + FNAC + TSH + surgery-select + histology
Papillary/medullary	Total + nodes + radioiodine + thyroxine
After	Calcium + voice + scar + follow-up + registry

⚠ FATAL TRAP

'Simple swelling' + no FNAC/TSH + missed cold/hoarse/nodes + quack-meds = cancer + cord-palsy + death! USG + FNAC + vocal + calcium + surgery + histology + audit.

★ NORCET HIT

HIT: Hashimoto + Graves + eye + storm + thyroid + pregnancy + neonatal + calcitonin + CEA + MEN + genes + counsel!

★ PNC PEARL: Cold, hoarse, nodal — needle it.

► Cold nodule + hoarse + nodes + 50 yrs. Ladder?

FNAC + scope + surgery + histology + MDT.

CBC: Hb, WBC, PLATELETS, PCV, ESR

Hb + TLC/DLC + platelets + PCV + MCV/MCH + ESR/CRP + smear + retics + trend + cause + treat + follow-up! Micro + macro + dimorphic + blasts + left-shift + eosinophilia + thrombocytosis + pancytopenia = marrow/bleed/hemolysis/infection hunt!

**Smear + trend
+ cause + treat**

**Micro/macro +
blasts + shift**

**Retic + LDH +
bilirubin + Coombs!**

▲ TEST — MEANS — ACT

Test	Means	Act
Hb/PCV/MCV	Anemia + type + cause	Iron/B12/folate + diet + recheck
WBC/DLC	Infection + blasts + shift	Culture + smear + marrow + senior
Platelets	Low/high + bleed/clot	Cause + guard + transfusion + HDU
ESR/CRP	Inflammation + trend + TB + follow-up	

⚠ FATAL TRAP

Numbers-without-smear + 'viral' + missed blasts/malaria + no trend + transfuse-every-low = mask + death + overload! Smear + trend + cause + treat + follow-up + audit.

★ NORCET HIT

HIT: Hypersegmented + macro-ovalocytes + B12/folate + NTD + nitrous + metformin + PPI + veg + levels + replacement!

★ PNC PEARL: Smear Chapter One, numbers Chapter Two.

► **Blasts + pallor + bleed + fever + child. Ladder?**

Flow + marrow + LP + risk + chemo + support.

COAGULATION TESTS: BT, PT/INR, aPTT

Bleed + clot + bruise + drug + liver + DIC + pre-op = BT + PT/INR + aPTT + platelets + fibrinogen + D-dimer + mixing + factors + bedside + senior! Prolonged + correction + trigger = factor/product + cause + HDU!

BT/PT/aPTT + platelets + site	Mixing + factors + bedside	Trigger + product + cause + HDU!
----------------------------------	-------------------------------	-------------------------------------

◆ TEST — MEANS — ACT

Test	Means	Act
BT/platelets	Platelet + vessel + von Willebrand	DDAVP + platelets + guard
PT/INR	Extrinsic + warfarin + liver + Vit K	Hold + Vit K + PCC + cause
aPTT	Intrinsic + heparin + hemophilia	Stop + factor + senior + HDU
DIC panel	Fibrinogen + D-dimer + trigger	Products + cause + ICU

⚠ FATAL TRAP

Heparin/warfarin-continue + 'oozing normal' + missed
DIC/HIT + no senior + blind-FFP = bleed + clot + death!
Hold + reverse + products + cause + HDU + audit.

★ NORCET HIT

HIT: HIT (heparin + platelets-fall + clot!) = stop +
argatroban + filter! Liver + Vit K + newborn (Vit K shot!)
+ pre-op + epidural + audit!

★ PNC PEARL: Mix it, map it, match product.

► Heparin + platelets-crash + clot + day-6. Name + swap?

HIT — stop + argatroban + filter + senior.

Group + Rh + crossmatch + label + bedside + vitals + reaction-kit + consent + indication + rate + monitor + document + handover! PRBC + platelets + FFP + cryo + albumin + leukoreduced + irradiated + CMV-safe + audit!

Group + cross + label + bedside	PRBC/platelet/FFP/ cryo + indication	Vitals + kit + monitor + audit!
------------------------------------	---	------------------------------------

Σ UNIT – USE – WATCH

Unit	Use	Watch
PRBC	Anemia/bleed + Hb + consent + rate	TACO/TRALI + K + calcium + vitals
Platelets	Low + bleed + dengue/chemo + ABO	Allergy + bacterial + count + rate
FFP/cryo	Coag + DIC + TTP + fibrinogen	Volume + allergy + TRALI + cause
Special	Leuko/irradiated/CMV + neonate + immune	

⚠ FATAL TRAP

Unlabeled + unchecked + fast + unwatched + 'one unit routine' + missed reaction = hemolysis + TRALI + death + suit! Label + bedside + slow + vitals + kit + stop + return + report.

★ NORCET HIT

HIT: Bombay + Rh-neg + rare + registry + autologous + cell-saver + restrictive + single-unit + recheck + stewardship + audit!

★ PNC PEARL: Right blood, right baby, right rate.

► **Transfusion + fever + chills + flank + red urine. Act trio?**

Stop + saline + return + workup + senior.

Glucose + urea/creatinine + LFT + lipids + electrolytes + ABG + HbA1c + thyroid + tumor-markers + therapeutic-drugs + trend + delta + panic + call + document + audit! Numbers + patient + trend + cause + treat, never alone!

**Trend + delta +
panic + call**

**Numbers + patient
+ cause + treat**

**Critical +
read-back + audit!**

★ TEST – PANIC – ACT

Test	Panic	Act
Glucose/K/ABG	Low/high + ECG + coma	Dextrose + Ca/bicarb + vent + ICU
Creatinine/LFT	Rise + oliguria + jaundice	Cause + fluids + drugs + HDU
Markers/drugs	Rise + toxic + subtherapeutic	Stage + dose + levels + senior
Trend	Delta + panic + call + document	Repeat + review + audit

⚠ FATAL TRAP

Numbers-alone + no trend/patient + missed panic + 'lab error' + no call = dead + suit! Repeat + correlate + call + read-back + document + senior + audit.

★ NORCET HIT

HIT: Hemolysis + lipemia + delay + fist-clench + tourniquet + posture + drugs + fasting + diurnal + reference + delta + audit!

★ PNC PEARL: Panic called, read back, written down.

► K critical + peaked + oliguria + ECG-change. Bundle?

Calcium + bicarb + dialysis + cause + ICU.

PLASMAPHERESIS (OVERVIEW)

Remove + replace + TTP/GBS/MG/crisis + ASFA + access + calcium + allergy + volume + audit!
Plasma out, albumin/FFP in — antibodies + complexes + toxins + paraproteins down, monitored + consented + scheduled!

**Out: antibodies +
complexes + toxins**

**In: albumin/FFP
+ monitored**

**Access + calcium
+ audit!**

▲ ITEM — RULE

Item	Rule
Indications	TTP + GBS + MG + vasculitis + select
Exchange	Access + volume + calcium + allergy + monitor
Replace	Albumin/FFP + consent + schedule + review
After	Labs + neuro + follow-up + registry + audit

⚠ FATAL TRAP

No consent/access-plan + calcium-missed + 'one session cures' + no schedule + lost follow-up = tetany + hypotension + relapse + suit! Consent + access + monitor + schedule + review.

★ NORCET HIT

HIT: ASFA categories + apheresis + LDL + liver + poisoning + transplant + neuro + renal + MDT + audit!

★ PNC PEARL: Out bad, in good, watch calcium.

► TTP + neuro + low-platelets + schistocytes. Lane?

Plasma-exchange + steroids + ICU + senior.

TRANSFUSION REACTIONS (LAB VIEW)

Stop + saline + vitals + return-bag + workup (hemolysis + TRALI + TACO + allergy + sepsis + hyperK!) + notify + document + audit! Fever/chills/flank/red-urine/dyspnea/rash/hypotension = REACT until proven otherwise!

**Stop + saline +
vitals + return**

**Hemolysis/TRALI/
TACO/sepsis hunt**

**Notify + document
+ audit!**

⚠ TYPE — CLUE — ACT

Type	Clue	Act
Hemolytic	Fever + flank + red + shock	Stop + fluids + workup + ICU
TRALI/TACO	Dyspnea + O2-drop + X-ray	Stop + O2 + sit + diuretics-TACO
Allergy/sepsis	Rash/wheeze + fever + rigors	Stop + epi/antihist + cultures
HyperK/overload	Peaked + JVD + crackles	Stop + Ca + slow + audit

⚠ FATAL TRAP

Continue + 'fever normal' + no return-bag/workup + rechallenge-same-unit = death + suit + repeat! Stop + saline + vitals + return + workup + notify + document + look-back.

★ NORCET HIT

HIT: DAT + eluate + culture-bag + chest + BNP + tryptase + IgA + washed + premedicate + slow + single-unit + stewardship!

★ PNC PEARL: Every chill stops the drip.

► **Dyspnea + O2-drop + X-ray-white + transfusion + no overload. Name + act?**

TRALI — stop + O2 + ICU + no-diuretics + report.